

Overview Document

Project: AIReason
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1

Background

The SFP study (System Frame Persistency) investigates the extent to which large language models maintain predefined decision frames stably across multiple steps.

The focus is not on the quality of individual responses, but on the structural capacity of models to consistently reproduce a priority logic once it has been established.

2

Objectives

The overall study examines three core aspects:

- **Persistence:** Do decision rules remain stable across sequences?
- **Drift:** When and how do deviations occur?
- **Priority Architecture:** How do models weight competing requirements?

The study does not follow a benchmark approach, but rather a behavioral-analytic perspective on LLM systems.

3

Study Structure

The SFP study consists of two interrelated sub-studies:

SFP-1 (Alpha / F-Test)

Focus: Fundamental capacity for frame stabilization under explicit conditions.

- Simple, clearly defined rules
- Direct instruction binding
- Baseline for persistence behavior

Objective: Determining whether models are capable of maintaining explicit frames in a stable manner.

SFP-2 (L-Study: L1–L3)

Focus: Variation of frame structure and its influence on decision stability.

- **L1** – Explicit rule (formal priority)
- **L2** – Semantic goal frame
- **L3** – Hierarchical priority structure

Objective: Analyzing how different frame formulations affect persistence and drift.

4

Shared Methodological Approach

All parts of the study follow a unified test logic:

- identical decision scenarios (8 items)
- binary decision structure (stable vs. short-term)
- multiple runs per model
- evaluation exclusively at the decision level

Metrics used:

- **Persistence Length (PL)**
- **First-Break Position (FBP)**
- **Persistence Rate (PR)**

Note: Justifications were not evaluated — only the decisions themselves.

5

Central Research Question

The overarching research question is:

→ *Under what conditions do LLMs interpret frames as binding rules — and when are they relativized or overridden?*

6

Meta-Findings (across SFP-1 and SFP-2)

The combined analysis reveals three fundamental patterns:

1. Explicit rules are unstable

→ Models rapidly lose formal specifications under generation pressure

2. Semantic frames are highly stable

→ Goal descriptions often exert stronger influence than explicit rules

3. Hierarchies are error-prone

→ Models remain stable, but partially follow incorrect priorities

7

Key Finding

The primary problem is not insufficient stability, but rather:

→ *stable, but displaced priority weighting*

This suggests that models possess implicit decision logics that can override explicit specifications.

8

Relationship Between Sub-Studies

SFP-1 and SFP-2 are not independent — they build upon each other:

- SFP-1 provides the baseline for frame stability
- SFP-2 shows how this stability changes across different frame types

Together, they constitute an investigation into how decision frames emerge, operate, and evolve in LLMs.

9

Classification

The SFP study does not primarily understand LLMs as text generators, but as:

→ *dynamic decision systems*

The findings are therefore relevant to:

- Alignment research
 - Agent design
 - Safe AI systems
 - Prompt and frame architectures
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Purpose of This Document

This document serves as a shared framework for:

- SFP-1 (Alpha / F-Test)
- SFP-2 (L1–L3)

It ensures that both studies remain:

- conceptually connected
- methodologically consistent
- jointly interpretable